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SAP Business One Integration Specification & M2M Developer Guide

Automated Komatsu Middle East Purchase Order Ingestion, Status Synchronization & Error Handling

1. Executive Summary & Architecture

This developer guide provides the authoritative Machine-to-Machine (M2M) API specifications for **ABS Consulting** to integrate **SAP Business One** with Dar Al Hai's **EQP System**. The EQP synchronization engine scans Komatsu Middle East's PDX portal 4 times daily (07:00, 11:00, 14:00, 18:00 Kuwait Time) and ingests confirmed Emergency Orders (EO), Daily Stock (DS), and Stock Orders (SO).

> **Core Integration Purpose:** ABS's integration service retrieves confirmed Komatsu orders in real time, automatically creates corresponding Purchase Orders (PO) in SAP Business One against Vendor `V000006` (Komatsu Middle East), and confirms PO creation back to EQP with the SAP `DocNum` and `DocEntry`.

Komatsu Ingestion Engine (EQP)

- Scans Komatsu PDX every 4 hours daily.
- Delta deduplication by SO and Quotation number.
- Delivery due date calculated per channel (EO: +30d, DS: +10d, SO: +120d).
- Automatic cookie health check & operator alerts.

SAP Business One Gateway (ABS)

- Polls `/orders/pending` or webhook pickup.
- Transitions orders to `PULLED_BY_SAP`.
- Inserts SAP B1 Purchase Order (DocType: Item).
- Acknowledges via `/orders/{id}/acknowledge`.

2. Hardened M2M Security & Authentication

Communication is restricted to HTTPS with strict Machine-to-Machine authentication. All integration requests must include a cryptographically hashed Bearer API key in the HTTP header:

```
Authorization: Bearer eqp_live_<32-byte-hex-secret>
```

- **Token Cryptography:** API keys are hashed with **SHA-256** prior to database persistence. Plaintext keys are never stored.
- **Scoped Authorization:** The integration service operates under the dedicated role `SAP_INTEGRATION_SERVICE` with granular scopes (`sap:read`, `sap:write`).
- **IP Whitelisting:** Endpoints enforce configurable IP whitelisting for ABS gateway IPs (e.g., Azure VM, AWS NAT gateway, or ABS static IP).
- **Rate Limiting:** Enforced at **60 requests per minute** per token to guarantee zero platform degradation.
- **Access Audit Trail:** Every request is logged in `api_access_logs` with response latency, client IP, payload, and HTTP status code.

3. Order Integration State Machine

State Enum	Trigger Event	Description & Action Required
PENDING_SAP_PULL	Komatsu Ingestion Cron	Order confirmed from Komatsu PDX. Ready for ABS/SAP pickup.
PULLED_BY_SAP	`GET /orders/pending`	Retrieved by ABS integration gateway. Locked against duplicate pickups.
SAP_PO_CREATED	`POST /acknowledge`	PO successfully created in SAP B1 with `sapDocNum` & `sapDocEntry`.
SAP_SYNC_FAILED	`POST /fail`	Integration failure reported by ABS (e.g., invalid item code). Flagged for retry.
EXCLUDED	Admin Action	Manually omitted by Dar Al Hai operations from SAP processing.

4. API Endpoints Specification

Method	Endpoint Path	Scope	Purpose
GET	`/health`	`sap:read`	Health & connectivity check; verifies credentials and latency.
GET	`/orders/pending`	`sap:read`	Retrieves pending confirmed orders (auto-transitions to `PULLED_BY_SAP`).
GET	`/orders/:id`	`sap:read`	Inspects single order manifest and line items by ID or SO number.
POST	`/orders/:id/acknowledge`	`sap:write`	Confirms SAP PO creation with `sapDocNum` & `sapDocEntry`.
POST	`/orders/:id/fail`	`sap:write`	Reports integration failure with detailed error diagnostic.

5. Request & Response Payloads

5.1 GET /orders/pending — Response Example


```
{
  "success": true,
  "totalCount": 1,
  "page": 1,
  "pageSize": 50,
  "returnedCount": 1,
  "orders": [
    {
      "id": "550e8400-e29b-41d4-a716-446655440000",
      "komatsuSalesOrderNo": "200451",
      "komatsuQuotationNo": "0000281449",
      "distributorRefNo": "R197/2026",
      "channel": "EMERGENCY_ORDER",
      "orderDate": "2026-10-12T07:00:00.000Z",
      "deliveryDueDate": "2026-11-11T07:00:00.000Z",
      "currency": "USD",
      "vendorCode": "V000006",
      "customerCode": "536K",
      "customerName": "DAR AL HAI",
      "equipmentSerial": "100433",
      "equipmentModel": "PC500LC-10R",
      "sapStatus": "PULLED_BY_SAP",
      "items": [
        {
          "lineNumber": 1,
          "partNumber": "6742-01-5280",
          "description": "GASKET, HEAD",
          "quantity": 2,
          "unitPriceUSD": 45.50,
          "lineTotalUSD": 91.00,
          "suggestedWarehouse": "003",
          "suggestedTaxCode": "P0"
        }
      ]
    }
  ]
}
```

5. Request & Response Payloads (Continued)

5.2 POST /orders/:id/acknowledge

Sent by ABS immediately after SAP B1 PO generation:

```
{
  "sapDocNum": "300481",
  "sapDocEntry": 1824,
  "remarks": "Created via ABS Integration Engine"
}
```

5.3 POST /orders/:id/fail

Sent by ABS when SAP B1 throws an error:

```
{
  "reason": "Item 6742-01-5280 not registered in Item Master",
  "errorCode": "-1002",
  "rawError": { "sapCode": -1002, "field": "ItemCode" }
}
```

6. SAP Business One Field Mapping Reference

EQP Field	SAP B1 Target Field (OPOR / POR1)	Type	Default / Transformation
vendorCode	OPOR.CardCode	String	Fixed: "V000006" (Komatsu Middle East)
orderDate	OPOR.DocDate	Date	Order timestamp
deliveryDueDate	OPOR.DocDueDate	Date	Calculated SLA due date (EO: +30d, DS: +10d, SO: +120d)
currency	OPOR.DocCur	String	Fixed: "USD"
distributorRefNo	OPOR.NumAtCard	String	Dar Al Hai Internal Ref (e.g., "R197/2026")
komatsuSalesOrderNo	OPOR.Comments	String	Stored in comments and UDF (e.g., U_KomatsuSO)
items[].partNumber	POR1.ItemCode	String	OEM Komatsu Part Number (e.g., "6742-01-5280")
items[].quantity	POR1.Quantity	Numeric	Line Item Quantity
items[].unitPriceUSD	POR1.Price	Numeric	Line Item Unit Price in USD
items[].suggestedWarehouse	POR1.WhsCode	String	Default: "003"
items[].suggestedTaxCode	POR1.TaxCode	String	Default: "P0" (0% Tax)

7. Implementation Code Examples for ABS Engineers

7.1 C# / .NET Example (HttpClient & SAP B1 DI API)

```
var client = new HttpClient();
client.DefaultRequestHeaders.Authorization = new AuthenticationHeaderValue("Bearer", "eqp_live_abs_secret_key");
const string baseUrl = "https://eqp.daralhai.com/api/v1/integration/sap";

// 1. Pull pending orders from EQP
var resp = await client.GetAsync($"{baseUrl}/orders/pending");
if (!resp.IsSuccessStatusCode) return;
var json = await resp.Content.ReadAsStringAsync();
using var doc = JsonDocument.Parse(json);

// 2. Iterate and insert Purchase Order into SAP B1 via DI API / Service Layer
foreach (var ord in doc.RootElement.GetProperty("orders").EnumerateArray()) {
    string orderId = ord.GetProperty("id").GetString();
    try {
        int sapDocNum = CreatePurchaseOrderInSap(ord); // Custom ABS method

        // 3. Acknowledge PO creation back to EQP
        var ack = new { sapDocNum = sapDocNum.ToString(), sapDocEntry = 1824, remarks = "Synced via ABS Service" };
        await client.PostAsJsonAsync($"{baseUrl}/orders/{orderId}/acknowledge", ack);
    } catch (Exception ex) {
        var fail = new { reason = ex.Message, errorCode = "-1001" };
        await client.PostAsJsonAsync($"{baseUrl}/orders/{orderId}/fail", fail);
    }
}
```

7.2 cURL CLI Verification Commands

```
# 1. Pull Pending Orders
curl -X GET "https://eqp.daralhai.com/api/v1/integration/sap/orders/pending"   -H "Authorization: Bearer eqp_live_abs_secr

# 2. Acknowledge SAP PO Creation
curl -X POST "https://eqp.daralhai.com/api/v1/integration/sap/orders/550e8400-e29b-41d4-a716-446655440000/acknowledge"   -
```

8. Verification & Testing Checklist

Initial Integration Test Procedure:

1. ABS receives initial sandbox/live token: `eqp_live_...`
2. Send `GET /health` to verify connectivity and IP whitelisting.
3. Send `GET /orders/pending` to verify JSON payload structure.
4. Execute test PO insertion in SAP B1 Test Company DB.
5. Submit acknowledgment to confirm automatic transition to `SAP_PO_CREATED`.

9. Production Contacts & Operations

Role	Contact Person	Email	Support Scope
Lead Systems Engineer	Motasem Ghanem	motasem.ghanem@daralhai.com	API Key generation, EQP logic, Swagger issues
ABS Integration Lead	Rand / ABS Team	abs-integration@daralhai.com	SAP B1 DI API, Service Layer mapping